Intro to commonly used pandas methods

DATA SCIENCE BOOTCAMP

NASHVILLE SOFTWARE SCHOOL

Objectives

1. Gain familiarity with the pandas library
2. Learn some common pandas methods

Introduction

Python is a general purpose programming language and is widely used for web applications, scientific computing, machine learning, and data analysis.

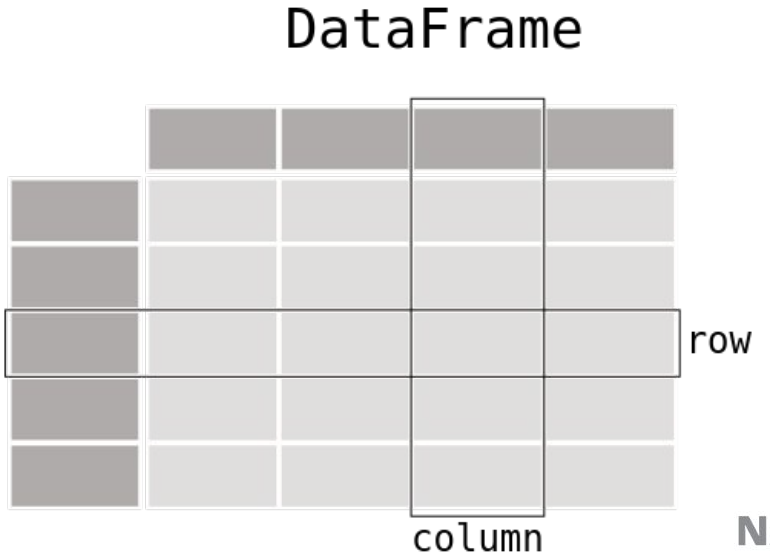
We'll be making use of the pandas library, an open source data analysis and manipulation tool.

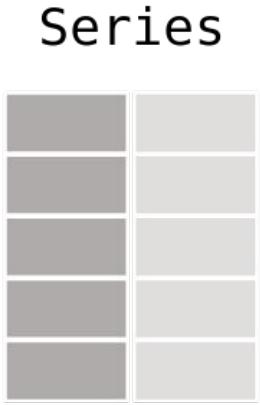
In Python, a library bundles together functions and objects that have a common functionality (like data analysis).

The pandas library was installed when you installed Anaconda.

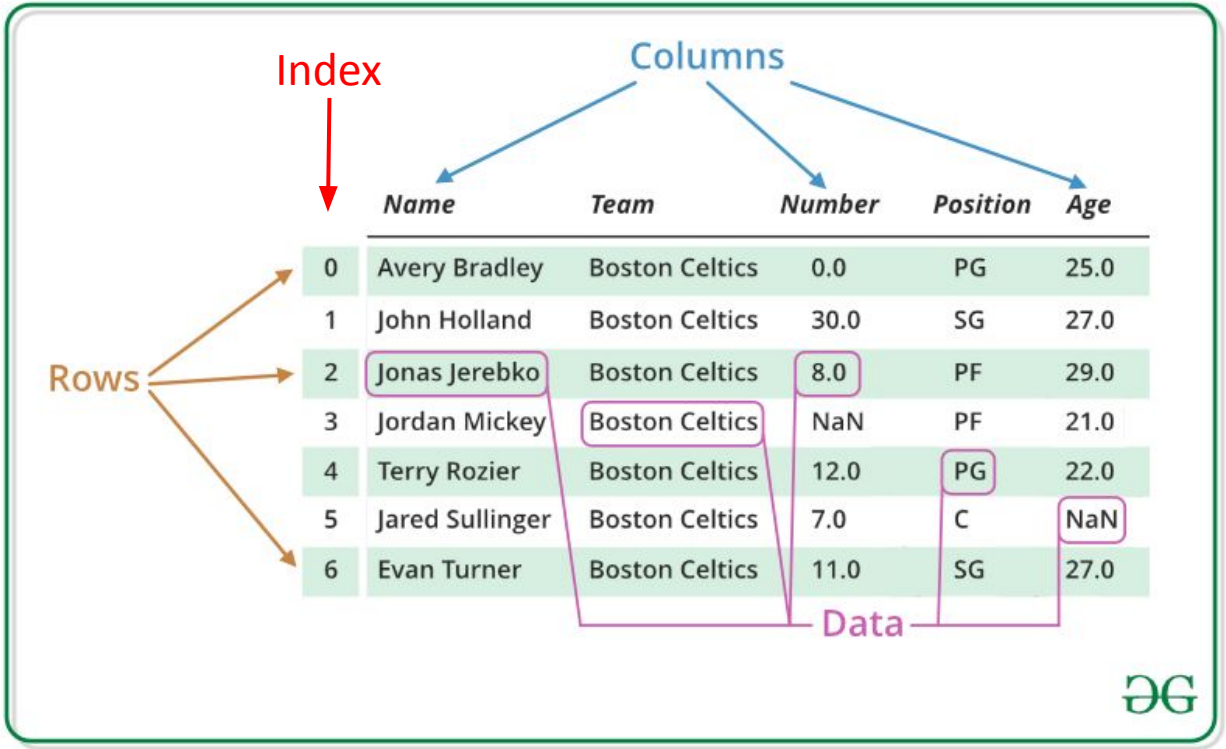
Dataframes and Series in pandas

The primary tool for working with data in *pandas* is the **DataFrame**, an object which can hold tabular data.
Each column of a DataFrame is a *pandas* **Series**.





Major Elements of a DataFrame



	Name	Team	Number	Position	Age
0	Avery Bradley	Boston Celtics	0.0	PG	25.0
1	John Holland	Boston Celtics	30.0	SG	27.0
2	Jonas Jerebko	Boston Celtics	8.0	PF	29.0
3	Jordan Mickey	Boston Celtics	NaN	PF	21.0
4	Terry Rozier	Boston Celtics	12.0	PG	22.0
5	Jared Sullinger	Boston Celtics	7.0	C	NaN
6	Evan Turner	Boston Celtics	11.0	SG	27.0

https://www.geeksforgeeks.org/python-pandas-dataframe/

To make use of *pandas*, we'll import it using the command

```
import pandas as pd
```

Allows us to use the components of *pandas.*

This is an **alias**, meaning that when referring to *pandas* objects, we will use the abbreviation *pd* instead of the full word.

Functions in pandas

Functions can be used as a way to direct Python to do certain actions, like read data in or modify a DataFrame.

When calling a function, you give the name of the function followed by a set of parentheses, which includes **arguments** that you need to pass in.

```
pd.read_csv('../data/public_art.csv')
```

```
pd.read_csv('../data/public_art.csv')
```

This indicates this function is part of the pandas library

```
pd.read_csv('../data/public_art.csv')
```

The name of the function is *read_csv*

This function reads in data from a **csv (comma separated values)** file into a *pandas* DataFrame

```
pd.read_csv('../data/public_art.csv')
```

We need parentheses since we're calling a function

```
pd.read_csv('../data/public_art.csv')
```

Here, we need an argument, which tells the function where to find our data

```
art = pd.read_csv('../data/public_art.csv')
```

We can **assign** the result of this function to a **variable** named art which will allow us to reuse it.

Methods

Functions can be part of a library (like *pd.read_csv()*) or they can be built into objects. When they are built into objects, we call them **methods**.

```
art.head()
```

This is the name of the DataFrame we are working with

*This function returns the first five rows of a DataFrame.

```
art.head()
```

This is the name of the method

```
art.head()
```

It does not require any arguments, but still needs parentheses since it is a function

Dataframes - Attributes / Properties

Dataframes also have additional characteristics called **attributes** or **properties** which are also accessed with a period but do not require parentheses.

For example, the *shape* attribute contains the dimensions (number of rows and number of columns) of a DataFrame.

```
art.shape
```

When initially reading in a dataset, there are several common operations that you'll use to gain familiarity with characteristics of the data that you're working with.

What follows next is a list of some of these commonly-used pandas tools.

Importing & Inspecting Data

Importing Data

- **pd.read_csv()** : Read in a comma delimited file
  - Good practice to look at the raw file in a text editor (like Visual Studio Code, ***not Excel***)
  - Additional arguments may be needed to handle extra rows at the top or extra data (footnotes) at the bottom

Inspecting

- **df.head() :** Look at the top rows of a DataFrame; 5 rows by default
- **df.tail() :** Look at the bottom rows of the DataFrame; 5 rows by default
- **df.shape** : Returns the dimensions of a DataFrame (# of rows, # of columns)
- **df.info():** Method to get information about the DataFrame

pandas documentation
https://pandas.pydata.org/pandas-docs/stable/reference/index.html

Modifying & Summarizing Data

Modifying

- **df.columns** : column labels attribute
- **df.rename()** : rename values
    - Can pass in a dictionary with existing columns as the key and new ones as the values
- **df.drop()** : drop the specified labels (either rows or columns) from the DataFrame

Summarizing

- **series.unique()** : returns the unique values in a column
- **series.nunique()** : returns the *number* of unique elements in a column
- **series.value_counts()** : returns the unique elements in a column and the number of appearances of each
- **series.describe()** : returns summary statistics of a numeric variable

pandas documentation
https://pandas.pydata.org/pandas-docs/stable/reference/index.html

Modifying & Summarizing Data

Slicing/Filtering

- **df.loc[]** – pass in row name and column name to access data at that location
- **df[[ ]]** - creates a slice (subset) of the DataFrame including just the columns passed in

*Notice that these last **properties** use square brackets instead of parentheses*

pandas documentation
https://pandas.pydata.org/pandas-docs/stable/reference/index.html